

BancaFiel Digital Transformation

AWS-Powered Loan Application Automation

Client: BancaFiel

Project: Automated Loan Processing System

Challenge: KPMG Lighthouse Innovation Competition

Team: [Your Team Name]

Date: February 15, 2026

The Opportunity

Transform a 1-week manual loan process into a 2-hour automated system using enterprise AWS services.

Business Impact: - \$264,625 monthly savings - 97% faster processing - 70% fraud reduction - 88,208% annual ROI

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Executive Summary

Problem: BancaFiel loses \$381,000 USD/month from a 1-week manual loan process driving customers to competitors.

Solution: AWS automation processes applications in 2 hours with 90% automation + human credit approval.

Impact: - Time: 1 week → 2 hours (97% reduction) - Savings: \$264,625 USD/month - ROI: 88,208% annually - Cost: \$300/month

Metric	Current	With AWS	Improvement
Processing Time	1 week	2 hours	97% faster
Application Loss	10%	3%	1,050 more approvals/month
Fraud Rate	2%	0.6%	70% reduction
Monthly Cost	\$381,070	\$116,445	\$264,625 saved

Problem Statement

Current Manual Process (15,000 applications/month)

1. Customer emails PDFs (INE/IFE, address proof, bank statements)
2. Staff downloads and manually types data into Excel (10-15 min/app)
3. Staff manually validates customer in database
4. Staff emails approver → waits 1-3 days
5. Approver reviews and decides
6. Staff updates ERP and emails customer

Total: 5-7 days per application

Business Impact

Monthly Losses: \$381,070 USD

Category	Cost	Cause
Lost Revenue	\$90,000	10% abandon (1,500/month) → switch to BBVA/Santander
Fraud Losses	\$283,500	2% fraud rate (189 cases/month) × \$1,500/case
Staff Costs	\$7,570	6-8 staff doing manual data entry

Competitive Gap: Modern banks approve in 2-4 hours. BancaFiel takes 1 week.

Our AWS Solution

Automated Workflow (Under 2 Hours)

1. Customer submits via web app (5 min)
2. **AWS Textract** extracts data from documents (30 sec)
3. **AWS Lambda** validates against database (10 sec)
4. **AWS Fraud Detector** checks fraud patterns (10 sec)
5. **Step Functions** routes to analyst by risk level (instant)
6. **Analyst reviews** pre-analyzed data → Approve/Reject (30 sec - 5 min)
7. System updates ERP + emails customer (instant)

Only manual step: Analyst clicks Approve/Reject. Everything else automated.

Core AWS Services

Service	Purpose	Impact
Textract	AI document reader	95-99% accuracy on Mexican IDs. 10 min → 30 sec
Lambda	Serverless processing	Auto-triggered functions. \$0.20 per 1M requests
RDS	Customer database	Replaces Excel. Millions of records
Fraud Detector	AI fraud prevention	2% → 0.6% fraud rate (saves \$198k/month)
Step Functions	Workflow orchestration	Visual workflow + approval routing
S3	Document storage	\$0.023/GB/month. Never lose documents
API Gateway	Security gate	Authentication + access control
Cognito	User authentication	Customer/staff separation

Architecture Flow

Customer → Web App → API → Lambda Functions

[S3] [Textract] [RDS] [Fraud Detector]

Step Functions

Route by Risk: High/Medium/Low

Analyst Dashboard

[Update ERP] [Send Email]

Cost Analysis

Current vs. AWS Costs

Category	Current	With AWS	Savings
Staff Labor	\$7,570	\$3,645	\$3,925
Lost Applications	\$90,000	\$27,000	\$63,000
Fraud Losses	\$283,500	\$85,500	\$198,000
AWS Operating	\$0	\$300	-\$300
TOTAL	\$381,070	\$116,445	\$264,625

AWS Operating Costs: \$300/month

Service	Monthly Cost
Textract (45k pages)	\$135
RDS (db.t3.small)	\$36
Fraud Detector	\$23
CloudWatch	\$15
S3, Amplify, Step Functions, Lambda	\$20
API Gateway, SES, Cognito	\$0 (free tier)
TOTAL	\$229 (~\$300 w/buffer)

ROI Summary

Metric	Amount
Monthly Savings	\$264,625 USD
Annual Savings	\$3,175,500 USD
Annual AWS Cost	\$3,600 USD
Annual ROI	88,208%
Payback Period	~4 days

\$1 spent on AWS = \$882 saved

Mexican Context

- Salaries: 2026 Mexican banking wages (Glassdoor/Indeed)
- Loan amounts: \$2k-5k average (Banco de México)
- Fraud rates: 2% (CNBV industry average)
- Conservative estimate range: \$142k-312k/month savings

Implementation Plan

4-Week Timeline

Week	Focus	Deliverable
1	Foundation: AWS setup, RDS/S3/Cognito, basic	Upload document → see

	React app, first Lambda	extracted data
2	Intelligence: Textract integration, test 50+ Mexican docs, fraud detection	Complete extraction + validation + fraud pipeline
3	Workflow: Step Functions, approval dashboard, email notifications	End-to-end workflow working
4	Polish: CloudWatch monitoring, testing (100+ apps), demo practice	Working demo ready for KPMG

Team Structure

Development (2-3): AWS infrastructure, Lambda functions, React frontend, database, testing

Business (2-3): Requirements, business case, project management, change management, presentations

Everyone: Learn AWS basics, understand system, practice demo

Key Risks

Risk	Mitigation
Textract accuracy low	Test 100+ Mexican docs Week 2. Add manual verification if <90%
Team availability	Agile approach, redistribute tasks
AWS costs exceed free tier	Monitor daily, use student credits, pool \$20-30 if needed
Demo failure	Record backup video, have screenshot slides
Can't finish in 4 weeks	MVP approach - show architecture + partial demo

Next Steps

This Week

- ☐ Team reviews briefing
- ☐ Prepare 7 questions for Session 2
- ☐ AWS account + student credits
- ☐ Assign roles
- ☐ Create workspace (GitHub, Drive)

Session 2 Questions (KPMG)

1. **Approval Criteria:** Credit score thresholds? Income/debt-to-income requirements?
2. **ERP System:** Current database? SQL-based? API access?
3. **Fraud Patterns:** Most common fraud types (forged docs, identity theft)?
4. **Document Formats:** PDF/images? Scanned vs. photos? Handwritten?
5. **Volume Patterns:** Peak times (month-end, paydays, seasonal)?

6. **Compliance:** Banking regulations (data retention, audit trails, security)?
7. **Staff Concerns:** Automation worries (job security, training)?

Goal: Gather specifics to refine solution

Final Deliverables (Week 7)

1. Live working demo
2. Business case presentation
3. Architecture diagrams
4. Updated cost/ROI analysis
5. Implementation roadmap
6. Change management strategy

Key Differentiators

Why We'll Win

1. **Real Working Prototype** - Functional AWS system, not PowerPoint
2. **Enterprise Technology** - Production-ready AWS services KPMG clients use
3. **Realistic Business Case** - Mexican salaries/fraud rates, conservative estimates
4. **Professional Approach** - Humans control credit decisions, comprehensive plan
5. **Complete Solution** - Technology + change management + implementation

Our Secret Weapon: Amazon Textract

vs. Low-Code Tools: - Most teams: N8N, Zapier, Airtable (\$30k-75k/month for 15k apps) - Us: Enterprise AI (\$300/month) - **100x cheaper + enterprise-grade**

Textract handles Mexican IDs with 95-99% accuracy - purpose-built for document extraction.

Why Act Now

Every month of delay = \$381,000 lost - 1,500 customers to competitors - 189 fraud cases - Staff burnout

3 months of "analysis" = \$1.14M lost

System pays for itself in 4 days.

Quick Reference

Key Metrics

Metric	Current	AWS	Change
Time	1 week	2 hours	-97%
Cost/month	\$381k	\$116k	-\$265k
Abandonment	10%	3%	-7%

Fraud	2%	0.6%	-70%
Staff	6-8	2-4	-50%

AWS Services Simplified

- **Textract** = AI reads PDFs (like Google Lens++)
- **Lambda** = Auto-triggered code (cloud Excel macros)
- **RDS** = Database (Excel for millions of rows)
- **S3** = Cloud storage (Google Drive for apps)
- **Step Functions** = Workflow manager (working flowchart)
- **Fraud Detector** = AI security guard

Resources

- AWS Training: aws.amazon.com/training/
- Student Credits: aws.amazon.com/education/awseducate/
- React: react.dev/learn

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Bottom Line: \$381k/month problem. \$300/month solution. That's our story.